1. DEFINITION

DEFINITION
Propylene glycol monolaurate (type I) EP/NF

DENOMINATION	
INCI NAME* (PCPC - International Nomenclature of Cosmetic Ingredients)	Propylene glycol laurate
UNII (Unique Ingredient Identifier) USP/FDA Substance Registration System (SRS)	40KT317HGP (Propylene glycol laurates) 668Z5835Z3 (Propylene glycol monolaurate)
*INCI name is given for information only. This ingredient is a pharmaceutical excipient. It must be used according to applicable regulations. Gattefossé accepts no responsibility for the use of this product in applications other than those recommended.	

2. FIELD OF USE

FIELD OF USE **This ingredient must be used according to appropriate regulations**
Pharmaceutical ingredient (oral, topical and/or rectal/vaginal routes)

3. MANUFACTURING PROCESS AND COMPOSITION

MANUFACTURING PROCESS
Esterification reaction between Propylene glycol and Lauric acid. No solvent or catalyst used during the manufacturing process.

COMPOSITION	
Well defined multi-constituent substance composed of Propylene glycol mono- and di- esters of Lauric acid (C12), the monoester fraction being predominant (Specification 45-70 %)	
Full Labeling (Key letters system) **$A_1 \geq 75.0$; $50.0 \leq A_2 < 75.0$; $25.0 \leq B < 50.0$; $10.0 \leq C < 25.0$; $5.0 \leq D < 10.0$; $1.0 \leq E < 5.0$; $0.1 \leq F < 1.0$; $G < 0.1$**	
Propylene glycol laurate	A1 ≥ 75.0%
Other constituents (i.e. incidental ingredients)	

GATTEFOSSÉ SAS
36 chemin de Genas - CS 70070 - 69804 Saint-Priest Cedex - FRANCE
Tel +33 4 72 22 98 00 - Fax +33 4 78 90 45 67 - www.gattefosse.com
SAS au capital de 4 650 000 Euros - R.C.S. Lyon B 389 586 900 - N° d'identité CEE FR 49 389 586 900

COMPOSITION
None

4. DRUG MASTER FILE

Country	Master file number
US DMF (Type IV – Excipient)	7818
Canada MF (Type III – Excipient)	/
For submission of the DMF in another country, please contact your local Gattefossé representative	

CHINA BUNDLING REVIEW			
(Generic name) Chinese name	**Registration N°**	**Company**	**CFDA/CDE Reference (http://www.cde.org.cn/yfb.do?method=main)**
(PROPYLENE GLYCOL MONOLAURATE TYPE I) 月桂酸丙二醇酯	F20210000305	Gattefossé SAS	Raw material has already been approved for use in marketed formulations in Joint-Review with drug applications (Status A)
Please contact our Regulatory Affairs department for further information and obtain letter of authorization			

5. PHARMACOPEIAS

CONFORMITY TO PHARMACOPEIAS	
Reference	**Conforms to monograph**
European Pharmacopoeia (Ph.Eur.) Current edition	Propylene glycol monolaurate (type I)
USP/ NF Current edition	Propylene glycol monolaurate (type I)
Japanese Pharmaceutical Excipient JPE Current edition	/
Chinese Pharmacopoeia Current edition	/

GATTEFOSSÉ SAS
36 chemin de Genas - CS 70070 - 69804 Saint-Priest Cedex - FRANCE
Tel +33 4 72 22 98 00 - Fax +33 4 78 90 45 67 - www.gattefosse.com
SAS au capital de 4 650 000 Euros - R.C.S. Lyon B 389 586 900 - N° d'identité CEE FR 49 389 586 900

6. VETERINARY MEDICINES FOR FOOD PRODUCING ANIMALS

EUROPEAN REGULATION Substances authorized to be used in		USA REGULATION Substances authorized to be used in			
Veterinary medicines and/or Biocidal products		Veterinary drugs		Pesticide products	
All products (except *)	* Products for use in food producing animals (husbandry)	All products (except *)	* Drugs for food producing animals (husbandry)	Applied to non-food use sites (e.g., flea and tick products in pets)	Applied to food use sites
✓ (9) warning: cat	✓ (1) All species (No limit)	✓ (9) warning: cat	✓ (5) USFA 172.856	✓ (5) 40CFR180.1250	✓ (5) 40CFR180.1250

EUROPEAN REGULATION
(1) Regulation (EU) No 37/2010 - Listed in annex II (Food additives: substances with a valid E number approved as additives in food stuffs for human consumption)
(2) Regulation (EU) No 37/2010 - Listed in annex II (Polyethylene glycol stearates with 8-40 oxyethylene units)
(3) Regulation (EU) No 37/2010 - Listed in annex II (Diethylene glycol monoethyl ether)
(4) EMA/CVMP - Substances considered as not falling within the scope of Regulation (EC) No 470/2009, with regard to residues of veterinary medicinal products in foodstuffs of animal origin
USA REGULATION
- Veterinary drugs
FDA is responsible for assuring that animal drugs are not only safe and effective for animals, but that food products from treated animals are safe for humans to consume. In the case of a drug for food producing animals, the inactive ingredients of the drug or its composition may be safe with respect to human food safety
(5) Approved food additives and GRAS substances for human food - Title 21, Part 172, 182—184 of the Code of Federal Regulations (CFR)
(6) Approved excipients (inactive ingredients) for use in human drugs by oral route (Ref: FDA Inactive ingredient database)
- Pesticides
(8)EPA's pesticides program (http://iaspub.epa.gov/apex/pesticides/f?p=INERTFINDER:1:0::NO:1::)

7. SPECIFIC PHARMACEUTICAL REGULATIONS

REGULATORY STATUS OF THE INGREDIENT FOR BORDER-LINE APPLICATIONS AUSTRALIA (TGA) - Australian Register of Therapeutic Goods Ingredients			
Status	**Ingredient Name**	**Use**	**Restrictions**
Approved	Propylene glycol monolaurate	Available for use as an Active Ingredient in: Biologicals, Prescription Medicines Available for use as an Excipient Ingredient in: Biologicals, Devices, Export Only, Listed Medicines, Over the Counter, Prescription Medicine	/

AUSTRALIA: Therapeutic goods are divided into 'medicines' and 'medical devices'. Some 'medicines' are limited to prescription-only while others are available without a prescription. Non-prescription medicines may be 'complementary medicines' (i.e. vitamin, mineral, herbal, aromatherapy, and homoeopathic products) or 'OTC medicines' and may be 'listed' (i.e. Sunscreens SPF4 and >4) or 'registered' in the ARTG.

REGULATORY STATUS OF THE INGREDIENT FOR BORDER-LINE APPLICATIONS NATURAL HEALTH PRODUCTS – CANADA NHPID		
Status	**NHPID name**	**Route of administration/ Restrictions**
Approved chemical name	Propylene glycol laurate	Role Non-medicinal: Purposes: Emulsifying Agent , Skin-Conditioning Agent - Emollient Acceptable Daily Intake Upper Limit: 25.0 Milligrams/kilogram body weight/day Route Of Administration: Oral Allowed Purposes: Emulsifying Agent General:

GATTEFOSSÉ SAS
36 chemin de Genas - CS 70070 - 69804 Saint-Priest Cedex - FRANCE
Tel +33 4 72 22 98 00 - Fax +33 4 78 90 45 67 - www.gattefosse.com
SAS au capital de 4 650 000 Euros - R.C.S. Lyon B 389 586 900 - N° d'identité CEE FR 49 389 586 900

REGULATORY STATUS OF THE INGREDIENT FOR BORDER-LINE APPLICATIONS NATURAL HEALTH PRODUCTS – CANADA NHPID		
Status	**NHPID name**	**Route of administration/ Restrictions**
		Route Of Administration: Topical Allowed Purposes: Skin-Conditioning Agent - Emollient Detail: Topical use only.
Under the Natural Health Products Regulations, which came into effect on January 1, 2004, natural health products (NHPs) are defined as: Probiotics; Herbal remedies; Vitamins and minerals; Homeopathic medicines; Traditional medicines such as traditional Chinese medicines; Other products like amino acids and essential fatty acids. The Natural Health Products Ingredients Database (NHPID) lists acceptable medicinal and non-medicinal ingredients used in Natural Health Products.		

8. SAFETY ASSESSMENT

SAFETY STUDIES PERFORMED BY GATTEFOSSE					
Study type/ duration	**Route**	**Species**	**Test article**	**Results/Conclusions**	**Reference**
Acute irritation (Patch test)	Dermal	Human	LAUROGLYCOL™ FCC pure (undiluted)	Cutaneous irritation index: 0.00 Very well Tolerated	(Gattefossé data, GAT -92210, 1992)
Acute irritation (OECD 404)	Dermal	Rabbit	LAUROGLYCOL™ FCC pure (undiluted)	Primary cutaneous irritation index: 1.08 Slightly Irritant	(Gattefossé data, GAT -90206, 1990)
Acute irritation (OECD 405)	Ocular	Rabbit	LAUROGLYCOL™ FCC pure (undiluted)	Mean ocular irritation index (48h): 0.33 Very slightly Irritant	(Gattefossé data, GAT -90206, 1990)
Acute toxicity (OECD 401)	Oral	Rat	LAUROGLYCOL™ FCC pure (undiluted)	$LD50_{(oral)}$ > 2006 mg/kg	(Gattefossé data, GAT -90207, 1990)
Acute toxicity (Dose escalating)	IV bolus (tail vein)	Mouse	LAUROGLYCOL ™ 90 Vehicle: Physiological saline solution M: 25, 50, 100, 200 and 150 mg/kg F: 25, 50, 100, 200, 400, 300, 250 and 200 mg/kg	$MTD_{(IV)}$ (M): 150 mg/kg $MTD_{(IV)}$ (F): 200 mg/kg	(Gattefossé data, GAT-MTDs-PH-08/0476, 2009)
Acute toxicity (eq OECD 423)	IV bolus (tail vein)	Mouse	LAUROGLYCOL™ 90 Vehicle: Physiological saline solution M+F: 150 mg/kg	$MTD_{(IV)}$: 150 mg/kg M+F: Administration at 150 mg/kg induces no mortality and toxicity signs.	(Gattefossé data, GAT-TAIs-PH-08/046, 2009)
Ames test (With and without activation) (OECD 471)	In vitro	S. typhimurium	LAUROGLYCOL ™ FCC pure	Negative. Not mutagenic.	(Gattefossé data, GAT-CIT 25742MMO, 2003)

GATTEFOSSÉ SAS
36 chemin de Genas - CS 70070 - 69804 Saint-Priest Cedex - FRANCE
Tel +33 4 72 22 98 00 - Fax +33 4 78 90 45 67 - www.gattefosse.com
SAS au capital de 4 650 000 Euros - R.C.S. Lyon B 389 586 900 - N° d'identité CEE FR 49 389 586 900

CIR COMPENDIUM (Cosmetic Ingredient Review) (http://www.cir-safety.org)			
Denomination	**Reference**	**Maximum "as used" concentration**	**Conclusions**
Propylene glycol mono and diesters	IJT, 18 (Suppl. 2) 35-52, 1999	Up to 50 %	Based on the available animal and clinical data included in this report, the CIR Expert Panel concludes that Propylene Glycol laurate is as cosmetic ingredients in the present practices of use.

EMA-Recommended Safety Limits (Excipients in the label and package leaflet of medicinal products for human use. EU Medicines Agency, CPMP/463/00 Rev. 1; publication planned end 2017)			
Propylene glycol esters	**Neonates up to 28 days (or 44 weeks post menstrual age for pre-terms)**	**1 month (29 days) up to 4 years**	**5 years up to 17 years and adults**
Safety limits	1 mg/kg	50 mg/kg	500 mg/kg

9. CHEMICAL REGULATIONS

CHEMICAL REGULATION			
INCI NAME	**CAS #**	**EINECS #**	**Chemical name**
Propylene glycol laurate	37321-62-3 (or 27194-74-7) (or 142-55-2)	253-462-2 (or 248-315-4) (or 205-542-3)	Dodecanoic acid, ester with 1,2-propanediol (or Dodecanoic acid, monoester with 1,2-propanediol) (or 2-hydroxypropyl laurate)

NATIONAL INVENTORIES					
INCI NAME	**U.S.A. (TSCA)**	**Canada (DSL/NDSL)**	**Australia (AIIC)**	**China (IECSC)**	**Other inventories**
Propylene glycol laurate	Not listed (or listed) (or listed)	Not listed (or DSL listed) (or Not Listed)	Not listed (or listed) (or listed)	Not listed (or Listed) (or Listed)	Dodecanoic acid, ester with 1,2-propanediol (EINECS, VNECI) (or Dodecanoic acid, monoester with 1,2-propanediol: AIIC, DSL, ENCS, IECSC, NZIoC, PICCS, TCSI, TSCA, VNECI, EINECS, AREC, ECL) (or 2-hydroxypropyl laurate: AREC, ECL, EINECS, AIIC, NZIoC, TCSI, VNECI)

U.S.A.: Substances that are regulated under other federal regulations, including food additives, drug, cosmetics… are not subject to TSCA regulation. "Naturally occurring substances" are not subject to Premanufacture Notice reporting.

Canada: Substances not listed on DSL or NDSL can be imported in Canada in quantities not exceeding 100 kg per 12-month period and per importer. Substances listed on NDSL only can be imported in Canada in quantities not exceeding 1000 kg per 12-month period and per importer. Please contact us if you go beyond these volumes. In addition, substances not listed in DSL or NDSL, but listed in the "in-commerce list" can be freely used in Products Regulated Under the Food and Drugs Act (F&DA). In addition, substances considered as "Natural" according Health Canada's definition are exempted from New Substances regulation.

Australia: The Australian Inventory of Chemical Substances (AICS) has been replaced by the Australian Inventory of Industrial Chemicals (AIIC) since July 2020. Chemicals listed on the AIIC are available for industrial use in Australia; however, importers must register with Australian authorities before importing the chemical in Australia. Some chemicals require specific information to be provided to Australian authorities.
Naturally occurring chemicals are exempted from AIIC listing; importers may have to submit a declaration.
Chemicals not listed on the AIIC must be assessed before importation.

GATTEFOSSÉ SAS
36 chemin de Genas - CS 70070 - 69804 Saint-Priest Cedex - FRANCE
Tel +33 4 72 22 98 00 - Fax +33 4 78 90 45 67 - www.gattefosse.com
SAS au capital de 4 650 000 Euros - R.C.S. Lyon B 389 586 900 - N° d'identité CEE FR 49 389 586 900

REACH REGULATION (REGISTRATION, EVALUATION, AUTHORISATION AND RESTRICTION OF CHEMICALS)		
	Status of the product	**Status per substance**
Propylene glycol laurate	Exempted	Substances intended to be used in medicinal products (human and/or veterinary use) are exempted from the main provisions on registration, evaluation and authorization, under REACh regulation.

10. HISTORY OF USE

10.1. Pharmaceutical applications

10.1.1. FDA IIG reference for Chemical equivalent

FDA INACTIVE INGREDIENT GUIDE (http://www.accessdata.fda.gov/scripts/cder/iig/index.cfm)				
Inactive ingredient	**Route**	**Dosage form**	**Maximum Potency per unit dose**	**Maximum Daily Exposure (MDE)**
No FDA IID reference exactly matches the definition propylene glycol monolaurate type I.				
Maximum potency and maximum daily exposure: Level info listed in the FDA IIG database refers to a specific product on the market. These levels do not constitute a regulatory max use level. When there is no info, the maximum potency per unit dose and Maximum Daily Exposure (MDE) fields will be blank.				

10.1.2. FDA IIG references to the most similar chemical (read-across approach)

In the **read-across approach**, endpoint information for one chemical (the source chemical) is used to predict the same endpoint for another chemical (the target chemical), which is considered to be "similar" in some way (usually on the basis of structural similarity or on the basis of the same mode or mechanisms of action). In principle, read-across approach can be used to assess physicochemical properties, toxicity, environmental fate and ecotoxicity.
For any of these endpoints, read-across approach may be performed in a qualitative or quantitative manner.'

The similarities may be based on the following:
- a common functional group (e.g. aldehyde, epoxide, ester, specific metal ion);
- common constituents or chemical classes, similar carbon range numbers;
- an incremental and constant change across the category (e.g. a chain-length category);
- the likelihood of common precursors and/or breakdown products, via physical or biological processes, which result in structurally similar chemicals.

These similarities justify data gap filling in a chemical category by applying read-across.

GATTEFOSSÉ SAS
36 chemin de Genas - CS 70070 - 69804 Saint-Priest Cedex - FRANCE
Tel +33 4 72 22 98 00 - Fax +33 4 78 90 45 67 - www.gattefosse.com
SAS au capital de 4 650 000 Euros - R.C.S. Lyon B 389 586 900 - N° d'identité CEE FR 49 389 586 900

FDA INACTIVE INGREDIENT GUIDE (http://www.accessdata.fda.gov/scripts/cder/iig/index.cfm)				
Inactive ingredient	**Route**	**Dosage form**	**Maximum Potency per unit dose**	**Maximum Daily Exposure (MDE)**
Propylene glycol monolaurate (Type II) (UNII: 668Z5835Z3)	ORAL	Capsule	NA	2115 mg
	ORAL	Tablet	NA	20 mg
	ORAL	Tablet, film coated	NA	20 mg
	TOPICAL	Gel	3% w/w	NA
	TRANSDERMAL	Film, extended release	NA	NA
Maximum potency and maximum daily exposure: Level info listed in the FDA IIG database refers to a specific product on the market. These levels do not constitute a regulatory max use level. When there is no info, the maximum potency per unit dose and Maximum Daily Exposure (MDE) fields will be blank.				

10.1.3. Other references

HANDBOOK OF PHARMACEUTICAL EXCIPIENTS (Current edition by Paul J. Sheskey, Bruno C Hancock, Gary P Moss Dabid J Goldfard)			
Denomination	**Functional Category**	**Regulatory status**	**Safety**
Propylene glycol monolaurate	Emollient Emulsifying agent Nonionic surfactant Oleaginous vehicule Penetration enhancer Solubilizing agent	Included in the FDA inactive Ingredients Database. Included in nonparenteral medicines licesed in the UK. Included in the Natural Health Products Ingredients Database.	Propylene glycol monolaurate is generally regarded as safe for use in pharmaceutical and cosmetic formulations in the present practices of use. Prolonged or frequently repeated skin contact may cause allergic reactions in hypersensitive individuals exposed to this product, and it may also cause eye irritation. LD50 (rat, oral) : 2.0 g/kg for lauroglycol 90 and lauroglycol FCC.

10.2. Food/Nutraceutical/Dietary supplement applications

Propylene glycol esters of Fatty acids have a long history of use worldwide as food additives for use in food, nutraceutical products and dietary supplements.

Country	Reference	Registration number
International	Evaluations Performed by Joint FAO/WHO Expert Committee on Food Additives	INS 477 Propane-1,2-diol esters of fatty acids
E.U.	EFA (European Food additives)	E 477 Propane-1,2-diol esters of fatty acids
U.S.A.	DSHEA (Dietary Supplement Health and Education Act) “Grandfathered ingredient” authorized for use in Dietary supplements	Esters of propylene glycol
U.S.A.	FCC (Food Chemical Codex)	Propylene glycol mono- and diesters

GATTEFOSSÉ SAS
36 chemin de Genas - CS 70070 - 69804 Saint-Priest Cedex - FRANCE
Tel +33 4 72 22 98 00 - Fax +33 4 78 90 45 67 - www.gattefosse.com
SAS au capital de 4 650 000 Euros - R.C.S. Lyon B 389 586 900 - N° d'identité CEE FR 49 389 586 900

Country	Reference	Registration number
U.S.A.	USFA US Food additives according to 21 CFR	21CFR §172.856 Propylene glycol mono-and diesters of Fats and Fatty acids
JAPAN	JSFA (Japanese Standards of Food Additives)	Propylene glycol esters of Fatty acids

HANDBOOK of FOOD ADDITIVES		
Denomination	**Use**	**Safety**
LAUROGLYCOL™ 90 Propylene glycol monolaurate	Food additive	FDA 21CFR §172.856; E 477; DMF no. 7818; worldwide food additive status

NB: The Food additive details are given for information. Gattefossé does not promote the use of this ingredient for Nutraceutical applications (food additive monograph conformity is not guaranteed). The suitability of the ingredient for use in nutraceutical products and dietary supplements remains the own responsibility of the company marketing the finished product.

GATTEFOSSÉ SAS
36 chemin de Genas - CS 70070 - 69804 Saint-Priest Cedex - FRANCE
Tel +33 4 72 22 98 00 - Fax +33 4 78 90 45 67 - www.gattefosse.com
SAS au capital de 4 650 000 Euros - R.C.S. Lyon B 389 586 900 - N° d'identité CEE FR 49 389 586 900